

# BAAVANSH REDDY GUNDLAPALLI

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## SUMMARY

Computer Science graduate from Rutgers University (Magna Cum Laude, 3.76 GPA) targeting backend and AI/ML engineering roles. Builds REST APIs, authentication systems, and data pipelines, with applied AI work in retrieval and embeddings — including AgentRecall, an open-source memory library for AI agents. Backed by a strong systems foundation across CPU simulators, compilers, and operating systems, built in C and Python. Personal portfolio includes an AI assistant recruiters can chat with to ask about this work.

## EDUCATION

**Rutgers University** — New Brunswick, NJ

May 2026

B.S. in Computer Science, Magna Cum Laude | GPA: 3.76 / 4.0 | Dean's List (4 semesters)

**Relevant Coursework:** Data Structures & Algorithms, Computer Architecture, Operating Systems, Compilers, Software Engineering, Databases, Artificial Intelligence

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, Java, C++, C, SQL

**Frameworks & Web:** Node.js, Express, FastAPI, React, Next.js, REST API design

**Databases & Storage:** PostgreSQL, MySQL, SQLite, MinIO (S3-compatible object storage)

**AI & ML:** RAG pipelines, vector search, embeddings, semantic search, PyTorch, NumPy, prompt engineering, LLM evaluation

**Tools & Concepts:** Git, Linux, unit testing (pytest, JUnit), object-oriented design, system design, Agile collaboration

## EXPERIENCE

**Backend Developer Intern** · Vellarro AI Research Center (Metasys Global)

Jan 2026 – Feb 2026

- Built the document-ingestion layer for an AI semantic-retrieval system, integrating MinIO (S3-compatible) object storage to programmatically extract heterogeneous, mixed-format files.
- Engineered the file-processing workflow that staged and prepared extracted documents for downstream chunking, embedding, and vector search.

**Full-Stack Engineering Extern** · Fidelis — LIVEY Event Platform

Jan 2026 – May 2026

- Co-built LIVEY, a full-stack event-discovery platform, by developing React search and filtering components and engineering REST API routes across the front and back end.
- Integrated Supabase (PostgreSQL) for real-time data storage, user authentication, and event management across the platform.

**Software Engineering Extern** · Little Red Riding Hood Inc.

Sep 2025 – Dec 2025

- Engineered secure authentication for a safety-focused rideshare platform, implementing role-based login and credential management for separate rider and driver accounts.
- Built driver session tracking and real-time geolocation handling that supported dispatch coordination and passenger-safety workflows.

**Python Programming Intern** · InternPe

Sep 2024 – Oct 2024

- Developed Python automation scripts for data processing and file handling, applying object-oriented design principles across structured engineering assignments.

## PROJECTS

## AgentRecall — Open-Source Memory Library for AI Agents | Python, SQLite, sentence-transformers, pytest

[github.com/BaavanshReddy/agentrecall](https://github.com/BaavanshReddy/agentrecall)

- Built and open-sourced AgentRecall, a ~500-line Python library that gives AI agents persistent memory using local SQLite storage and on-device embeddings, with no cloud database or API keys required.
- Implemented semantic retrieval with cosine similarity and a Maximal Marginal Relevance (MMR) diversity mode, plus per-agent namespacing, and verified the public API with 15 passing unit tests.

## Chat-with-Baavansh — AI Portfolio with Embedded RAG Assistant | Next.js, TypeScript, Claude API [baavansh-portfolio.vercel.app](https://baavansh-portfolio.vercel.app)

- Built a personal portfolio site with an embedded RAG assistant that answers recruiter questions from a structured knowledge base, pairing a Claude-powered engine with an automatic in-browser fallback.

## RISC-V CPU Simulator | C, Computer Architecture

[github.com/BaavanshReddy/risc-v-simulator](https://github.com/BaavanshReddy/risc-v-simulator)

- Implemented a single-cycle RISC-V CPU simulator with full instruction decode and datapath execution, backed by a 1KiB direct-mapped cache with hit/miss tracking to model a realistic memory hierarchy.

## TinyL Compiler | Python, Compilers

[github.com/BaavanshReddy/tinyl-compiler](https://github.com/BaavanshReddy/tinyl-compiler)

- Built a complete compiler — lexer, recursive-descent parser, AST, intermediate representation, and code generation — translating the TinyL language from a formal grammar to stack-based bytecode.

## RuPizza — Object-Oriented Ordering Application | Java, JavaFX, JUnit

[github.com/BaavanshReddy/RUPizza](https://github.com/BaavanshReddy/RUPizza)

- Developed a JavaFX desktop ordering application designed with object-oriented principles — inheritance, polymorphism, and the Factory pattern — featuring an order builder, store-order management, and JUnit test coverage.

## Neural Network from Scratch | Python, NumPy, PyTorch

[github.com/BaavanshReddy/llm-digit-face-training](https://github.com/BaavanshReddy/llm-digit-face-training)

- Implemented a perceptron and a 3-layer neural network with hand-coded forward propagation and backpropagation, reaching 89.1% face-recognition and 86.8% digit-classification accuracy validated against a PyTorch baseline.

## Linux Filesystem Simulator | C, Operating Systems

[github.com/BaavanshReddy/linux-filesystem](https://github.com/BaavanshReddy/linux-filesystem)

- Engineered a Unix-style virtual filesystem in C11 — inode-based storage, rwx permission management, and binary serialization for persistence — exposing 15+ shell commands through an interactive REPL and verified leak-free under Valgrind and AddressSanitizer.

## Campus Event Scraper | Python, BeautifulSoup, SQLite

[github.com/BaavanshReddy/campus-event-scraper](https://github.com/BaavanshReddy/campus-event-scraper)

- Built a Python web-scraping pipeline that collects and normalizes university event listings into an indexed SQLite database, staying reliable through exponential-backoff retries, URL-based deduplication, and graceful offline fallback when sources are unavailable.

## RESEARCH

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### LLM FactCheck — Factual QA Evaluation Framework | Python, RAG, BM25, TriviaQA

- Co-authored a research study benchmarking LLM factual question-answering across direct prompting, BM25 retrieval, and RAG on a controlled 100-question TriviaQA evaluation set.
- Found that BM25 surfaced supporting evidence in 90% of cases while RAG improved exact-match accuracy by only 3 points (0.66 to 0.69), isolating generation quality — not retrieval — as the bottleneck.

## LEADERSHIP & ACTIVITIES

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- **Community Service Officer, Rutgers University Police Department** (Jan 2025 – May 2026) — Supported incident response and crowd management for large-scale university events.
- **Leadership Roles, Alpha Phi Delta Fraternity** (May 2024 – May 2026) — Held three elected positions (Risk Management Chair, Head of Judiciary Board, Co-Brotherhood Chair); led safety compliance, conduct governance, and member engagement.
- **Administration Secretary, ENIGMA Technical Society** (Aug 2023 – May 2024) — Organized technical workshops and streamlined communication across society events.

## ADDITIONAL

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**Languages:** English, Hindi, Telugu

**Activities:** Former NCAA Division I Crew Athlete, Rutgers University